

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU
July / August 2019 Supplementary Examinations

Programme: B.E.

Branch : Information Science and Engineering

Course Code: 15IS4DCADA

Course: Analysis and Design of Algorithms

Semester : IV

Duration: 3 hrs.

Max Marks: 100

Date: 29.07.2019

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may suitably assumed.

UNIT - I

- 1 a) Describe with a neat flow chart, the process of algorithm design and analysis. **08**
- b) Prove that the running time of recursive algorithm used to solve Towers of Hanoi problem is exponential. **08**
- c) Is selection sort a stable sorting algorithm? Prove your answer with an example. **04**

UNIT - II

- 2 a) Sort the list {Q, U, E, S, T, I, O, N} in alphabetical order using Bubble sort algorithm. Prove that if bubble sort makes no exchanges on its pass through a list, the list is sorted and the algorithm can be stopped. **07**
- b) Given two sorted arrays $A[p \dots q]$ and $B[q+1 \dots r]$, write an algorithm to merge these arrays into one sorted array $C[p \dots r]$. Analyze the running time of the algorithm designed. **06**
- c) Apply divide and conquer design strategy to multiply two integers: 1629 and 7328 **07**

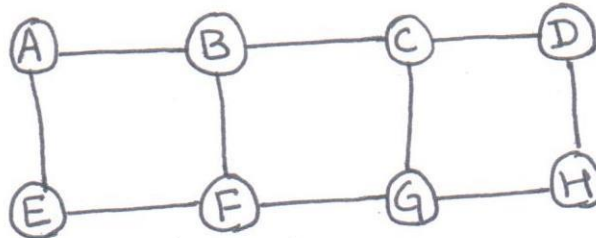
OR

- 3 a) Write the brute-force string matching algorithm. Calculate the number of character comparisons in the worst case and give an example for such a worst case input. **08**
- b) Demonstrate of how quicksort works on the input dataset: 5,4,9,7,1,8,3,2,6. Draw the tree of recursive calls made. **08**
- c) Design a divide and conquer based approach to compute x^{62} with only 9 multiplications. **04**

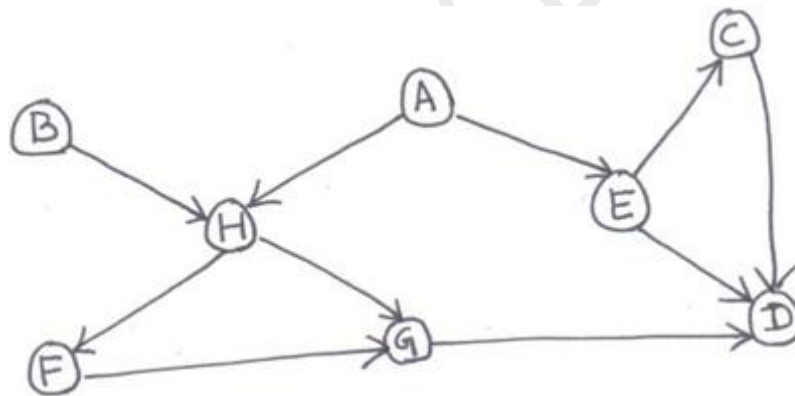
Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

UNIT - III

- 4 a) Design a suitable algorithm for finding a minimum edge path between two given vertices in any given graph. Apply that algorithm to the graph below showing the tree that identifies the minimum edge path from A to G. 08



- b) Construct a 2-3 tree for the list: 10,6,9,4,3,5,8 06
- c) Solve the topological sorting problem for the given graph 06



OR

- 5 a) Differentiate between BFS and DFS algorithms. 04
- b) Given an array of 6 elements: 15,19,10,7,17,16, sort it in ascending order using heapsort. 08
- c) What is an AVL tree? Explain the four types of rotations used to construct the AVL tree. 08

UNIT - IV

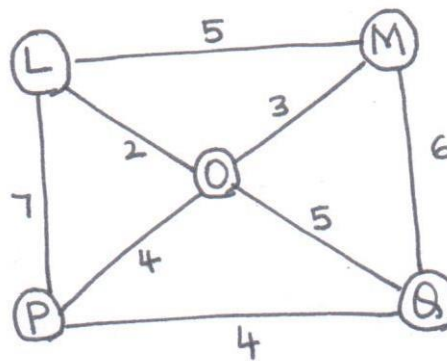
- 6 a) Using Horspool algorithm, search for the pattern 'BAOBAB' in the text: BESS_KNEW_ABOUT_BAOBABS 07
- b) Write the pseudo code which implements computation of binomial coefficient using dynamic programming. 06

- c) Apply bottom up dynamic programming algorithm to the following instance of knapsack problem. Capacity of the knapsack is 6. **07**

Item	Weight	Value
1	3	25
2	2	20
3	1	15
4	4	40
5	5	50

UNIT - V

- 7 a) Apply Prim's algorithm for the following graph to find the Minimum Spanning Tree. **07**



- b) Construct Huffman code tree for the following data. Encode the text ADDBEC **08**

Character	A	B	C	D	E
Probability	0.2	0.3	0.11	0.27	0.12

- c) Define P class and NP class of problems with an example for each. **05**
